

Test 36: Byte-level robustness

Verdict: PASS | Generated: 2026-09-02 23:33:00 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Mixed: $|r|$ under 256-level quantization + RNG entropy (statistical). Sample accumulated to 25 000 (pass 1) / 50 000 (pass 2) eigenvalues; verdict = multi-window average over 120 windows

METHOD: Mixed: $|r|$ under 256-level quantization + RNG entropy (statistical). Sample accumulated to 25 000 (pass 1) / 50 000 (pass 2) eigenvalues; verdict = multi-window average over 120 windows

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik}^2 + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \cdot W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

$q \cdot W \ll t$ (e.g. $W=4$, $q=0.3 \Rightarrow V \approx 1.2$ next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and r drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (N_v ,

layout, charges) of that realization, not on the AB physics.

$q \cdot W \gg t$ ($W \gg 1$): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$:

they certify the PHASE physics, not the disorder physics.

Sample: 55290 eigs (pass 2 (secondary), target 50000, reached=yes); $W_{\text{eff}}=1.00$.

Byte robustness verdict = multi-window $|r| = 0.0009$ (tol 0.0200, 120 windows); single-window $|r| = 0.0061$ (diagnostic).

Test 36: byte robust ? $r_{256}=0.5756$, $|r|_{\text{multi}}=0.0009$ (120 windows), $|r|_{1\text{win}}=0.0061$ (diag), $H=7.9991$ bits, $z=-0.234$, $n=55288$? PASS

(no plots queued by this pass ? physics series live in the HARDCORE pass-2 report: hc_verdict_table queues a sub-check verdict map there, and the deep-audit tests queue their own)

Result

Reproduce with: `julia ab_cloud_v19.jl --test 36 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Plots

Independent verification guide

FULL COMPUTATION LOG

[22:45:53.699] [+32268.981s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[22:51:13.311] [+32588.593s] t37 pass 2 (secondary) realization 1: central block 5529 eigs, accumulated 5529 (target 50000)

[22:51:14.494] [+32589.776s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[22:55:54.778] [+32870.060s] t37 pass 2 (secondary) realization 2: central block 5529 eigs, accumulated 11058 (target 50000)

[22:55:56.081] [+32871.363s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[23:00:33.946] [+33149.228s] t37 pass 2 (secondary) realization 3: central block 5529 eigs, accumulated 16587 (target 50000)

[23:00:35.139] [+33150.422s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[23:05:51.910] [+33467.192s] t37 pass 2 (secondary) realization 4: central block 5529 eigs, accumulated 22116 (target 50000)

[23:05:53.552] [+33468.834s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[23:10:20.477] [+33735.759s] t37 pass 2 (secondary) realization 5: central block 5529 eigs, accumulated 27645 (target 50000)

[23:10:21.619] [+33736.901s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[23:14:36.092] [+33991.374s] t37 pass 2 (secondary) realization 6: central block 5529 eigs, accumulated 33174 (target 50000)

[23:14:37.266] [+33992.548s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[23:18:57.442] [+34252.724s] t37 pass 2 (secondary) realization 7: central block 5529 eigs, accumulated 38703 (target 50000)

[23:18:58.673] [+34253.955s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[23:23:11.865] [+34507.147s] t37 pass 2 (secondary) realization 8: central block 5529 eigs, accumulated 44232 (target 50000)

[23:23:13.003] [+34508.286s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[23:27:55.567] [+34790.849s] t37 pass 2 (secondary) realization 9: central block 5529 eigs, accumulated 49761 (target 50000)

[23:27:56.843] [+34792.126s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[23:33:00.249] [+35095.531s] t37 pass 2 (secondary) realization 10: central block 5529 eigs, accumulated 55290 (target 50000)

[23:33:00.262] [+35095.544s] calc: ?r? = 0.387930 (n_r=55288, n_eigs=55290) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.277] [+35095.560s] calc: ?r? = 0.657053 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.560s] calc: ?r? = 0.643724 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.560s] calc: ?r? = 0.624209 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.560s] calc: ?r? = 0.568482 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.560s] calc: ?r? = 0.522107 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.560s] calc: ?r? = 0.535764 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.652837 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.595610 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.624374 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.604567 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.485897 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.693192 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.683830 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.528901 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.278] [+35095.561s] calc: ?r? = 0.519134 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.492447 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.622341 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.617574 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.607611 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.545788 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.540762 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.384361 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.605531 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.527459 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.492429 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.640946 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: $\chi^2 = 0.815854$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.467241 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.655008 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.651800 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.279] [+35095.561s] calc: ?r? = 0.476787 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.522209 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.526679 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.487008 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.639582 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.663925 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\chi^2 = 0.501332$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.462589 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.486558 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.571701 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\chi^2 = 0.655437$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\chi^2 = 0.669788$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\chi^2 = 0.681240$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.655458 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: ?r? = 0.628425 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\chi^2 = 0.585196$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\tau = 0.563536$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\tau = 0.692005$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\tau = 0.726317$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\tau = 0.665739$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\tau = 0.612582$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.562s] calc: $\tau = 0.672288$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.537854$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.462356$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.634012$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.544264$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.407715$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.540566$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.458490$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.572290$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.612044$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.280] [+35095.563s] calc: $\tau = 0.745189$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.281] [+35095.563s] calc: $\tau = 0.606921$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.281] [+35095.563s] calc: $\tau = 0.673025$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.281] [+35095.563s] calc: $\tau = 0.399152$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.281] [+35095.563s] calc: $\tau = 0.553205$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.281] [+35095.563s] calc: $\tau = 0.548260$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.281] [+35095.563s] calc: $\tau = 0.521588$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.281] [+35095.563s] calc: $\tau = 0.546769$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.533360$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.501554$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.508977$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.610973$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.626084$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.550313$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.550351$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.642275$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.529751$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.548388$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: $\tau = 0.658126$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.557107 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.636281 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.497804 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.684932 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.687670 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.570463 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.610529 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.596610 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.617638 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.380672 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.525082 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.596661 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.578898 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.512743 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.628458 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.650209 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.535112 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.599149 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.454103 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.592745 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.490758 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.482994 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.451370 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.611679 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.530496 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.282] [+35095.564s] calc: ?r? = 0.515716 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.565s] calc: ?r? = 0.547860 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.565s] calc: ?r? = 0.699975 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.565s] calc: ?r? = 0.619194 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.565s] calc: ?r? = 0.372409 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.565s] calc: ?r? = 0.670355 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.565s] calc: ?r? = 0.708571 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.607556 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.621021 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.662679 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.484236 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.642217 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.541220 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.602689 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.479526 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[23:33:00.283] [+35095.566s] calc: ?r? = 0.599213 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

CONSOLE CAPTURE

??

TEST 36: Byte-level robustness ? quantization + RNG entropy

Verify ?r? survives 256-level quantization + RNG byte uniformity

??

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

V_i = ?_k q_k•W / (r_ik?•N?, + 1), hopping t = 1.

The ratio (disorder energy q•W) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

? q•W ? t (e.g. W=4, q=0.3 ? V?1.2 next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and ?r? drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (Nv,

layout, charges) of that realization, not on the AB physics.

? q•W ? t (W ? 1): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at W_eff = min(cfg.ab_W, W_max = 1.0):

they certify the PHASE physics, not the disorder physics.

Sample target: 50000 eigenvalues [pass 2 (secondary)] (cfg.ab37_n_eigs_pass2)

Disorder: W = 1.00 = min(ab_W=4.00, W_max=1.00) ? q•W = 1.00 vs hopping t = 1.0 ? AB phases dominate (byte check tests the phase physics)

realization 1/5, continuing (below target): +5529 central eigs ? accumulated 5529 / 50000 (11%)

realization 2/5, continuing (below target): +5529 central eigs ? accumulated 11058 / 50000 (22%)

realization 3/5, continuing (below target): +5529 central eigs ? accumulated 16587 / 50000 (33%)

realization 4/5, continuing (below target): +5529 central eigs ? accumulated 22116 / 50000 (44%)

realization 5/5, continuing (below target): +5529 central eigs ? accumulated 27645 / 50000 (55%)

realization 6/6, continuing (below target): +5529 central eigs ? accumulated 33174 / 50000 (66%)

realization 7/7, continuing (below target): +5529 central eigs ? accumulated 38703 / 50000 (77%)

realization 8/8, continuing (below target): +5529 central eigs ? accumulated 44232 / 50000 (88%)

realization 9/9, continuing (below target): +5529 central eigs ? accumulated 49761 / 50000 (100%)

realization 10/10: +5529 central eigs ? accumulated 55290 / 50000 (111%)

Raw $r = 0.3879$ (n=55288 spacings)

Universality class: Poisson (d_GUE=0.2117, d_GOE=0.1428, d_GSE=0.2883, d_Poi=0.0016)

(local window: n=15 central eigs, raw $r_{\text{local}} = 0.6571$)

(a) Byte-quantization robustness (256 levels):

legacy single window (diagnostic): $r_{\text{byte}} = 0.6631$, $r_{\text{local}} = 0.6571$, $|r| = 0.0061$

multi-window average (decisive; 120 windows $\times$ 15 eigs):

$r_{\text{raw}} = 0.5765$, $r_{\text{byte}} = 0.5756$, $|r| = 0.0009$ (tol=0.0200) ? PASS

(b) RNG byte-entropy / uniformity:

Shannon entropy = 7.9991 bits (target ? 7.9)

z (CLT approx, df=255) = -0.2342 ($|z| < 3.0$) ? PASS

Test 36: byte robust ? $r_{256} = 0.5756$, $|r|_{\text{multi}} = 0.0009$ (120 windows), $|r|_{1\text{win}} = 0.0061$ (diag), H=7.9991 bits, $z = -0.234$, n=55288 ? PASS